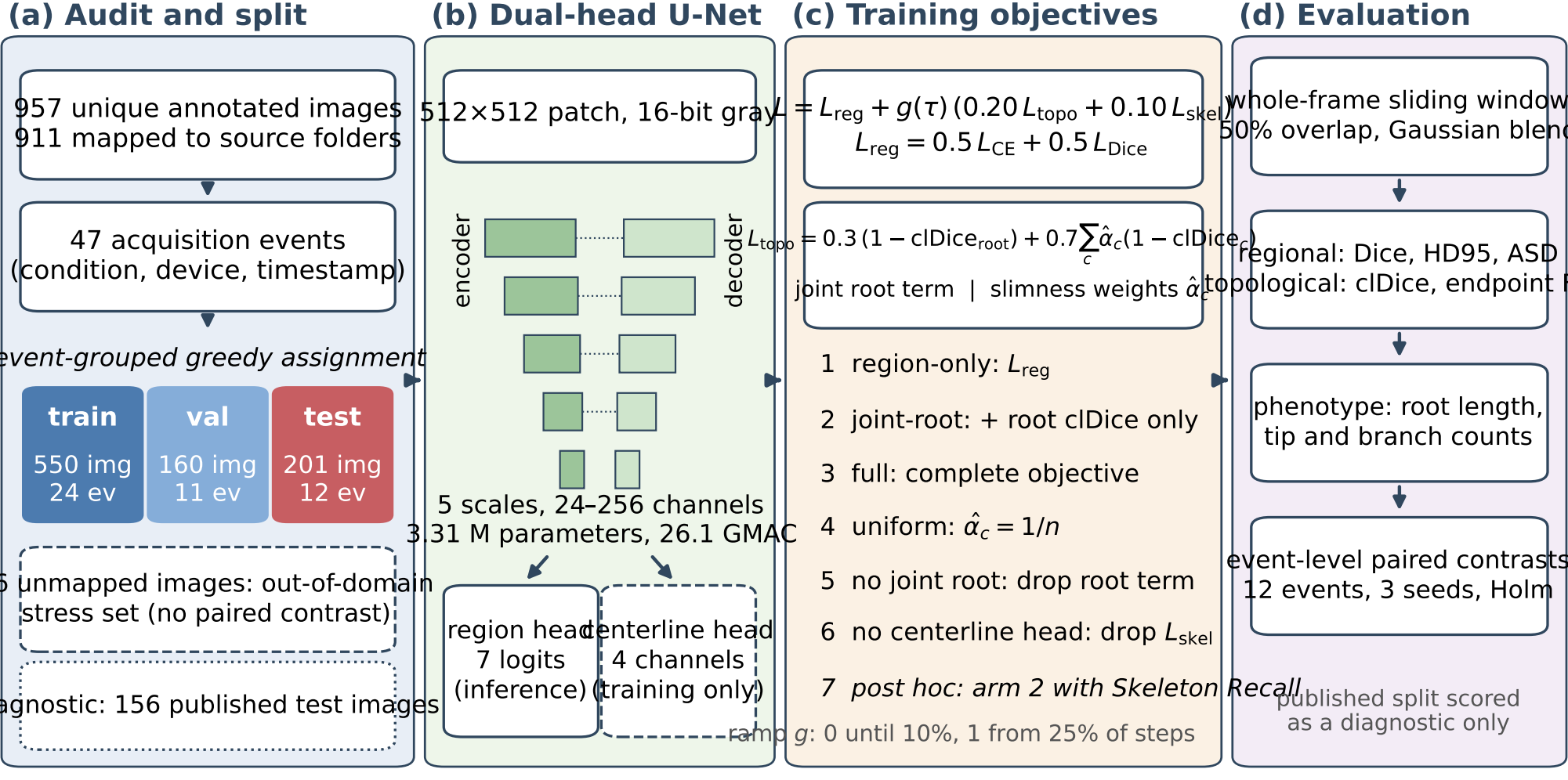
(a) Audit and split
957 unique annotated images
911 mapped to source folders
47 acquisition events
(condition, device, timestamp)
event-grouped greedy assignment
train
550 img
24 ev
val
160 img
11 ev
test
201 img
12 ev
6 unmapped images: out-of-domain
stress set (no paired contrast)
agnostic: 156 published test images
(b) Dual-head U-Net
512×512 patch, 16-bit gray
encoder
decoder
5 scales, 24–256 channels
3.31 M parameters, 26.1 GMAC
region head
7 logits
(inference)
centerline head
4 channels
(training only)
(c) Training objectives
$L = L_{reg} + g(\tau)\,(0.20\,L_{topo} + 0.10\,L_{skel})$
$L_{reg} = 0.5\,L_{CE} + 0.5\,L_{Dice}$
$L_{topo} = 0.3\,(1 - \mathrm{clDice}_{root}) + 0.7\sum_c \hat{\alpha}_c(1 - \mathrm{clDice}_c)$
joint root term | slimness weights $\hat{\alpha}_c$
1 region-only: L_{reg}
2 joint-root: + root clDice only
3 full: complete objective
4 uniform: $\hat{\alpha}_c = 1/n$
5 no joint root: drop root term
6 no centerline head: drop L_{skel}
7 post hoc: arm 2 with Skeleton Recall
ramp g: 0 until 10%, 1 from 25% of steps
(d) Evaluation
whole-frame sliding window
50% overlap, Gaussian blend
regional: Dice, HD95, ASD
topological: clDice, endpoint
phenotype: root length,
tip and branch counts
event-level paired contrasts
12 events, 3 seeds, Holm
published split scored
as a diagnostic only